2.4 Nitrogen & Sulfur

Question Paper

Course	CIE A Level Chemistry
Section	2. Inorganic Chemistry
Topic	2.4 Nitrogen & Sulfur
Difficulty	Hard

Time allowed: 40

Score: /28

Percentage: /100

Question 1a

The combustion of fuels in motor vehicles, trains, aeroplanes and power stations produces the pollutant gas NO_2.

Write an equation to show how NO_2 is formed in these situations.

[1 mark]

Question 1b

i)
State how NO_2 is removed from the exhaust gases of motor vehicles.

[1]

ii)
Write an equation for this process.

[1]

[2 marks]

Question 1c

Suggest whether the production of the pollutant NO_2 would be reduced if fossil fuels were replaced by hydrogen as a fuel for combustion. Explain your answer.

[1 mark]

Question 1d

In the atmosphere, NO_2 acts as a catalyst for the oxidation of SO_2 to SO_3.

$$SO_2(g) + \frac{1}{2} O_2(g) \xrightarrow{NO_2} SO_3(g)$$

i)
What is the environmental significance of this reaction?

[1]

The oxidation takes place in two steps. The initial reaction is between NO_2 and SO_2.

reaction **1** $NO_2(g) + SO_2(g) \rightleftharpoons NO(g) + SO_3(g) \quad \Delta H = -168\ \text{kJ mol}^{-1}$

ii)
Write an equation to show how the NO_2 is regenerated in the second step of the oxidation.

[1]

iii)
The temperature of the atmosphere decreases with height. Explain how this will affect the position of the equilibrium in reaction **1**.

[2]

[4 marks]

Question 2a

This question is about ammonia.

i)
Explain how ammonia can be prepared from ammonium chloride and calcium hydroxide.

[2]

ii)
Explain how the presence of ammonia gas can be confirmed.

[1]

[3 marks]

Question 2b

Explain why the reaction between ammonium chloride and calcium hydroxide is an example of an acid-base reaction.

[2 marks]

Question 2c

i)

Draw a diagram of the ammonium ion showing all of the bonds within the molecule.

[2]

ii)

The N-H bonds in the ammonium ion have been found to be equal in length. Suggest what this means about the bonds or charge on the ion.

[1]

[3 marks]

Question 3a

A molecule of nitrogen will only react with oxygen under extreme conditions due to the strong triple bond between nitrogen atoms.

i)
State another reason why the nitrogen molecule is so unreactive.

[1]

ii)
State the type of covalent bonds that exists between the two nitrogen atoms.

[2]

[3 marks]

Question 3b

Despite the unreactivity of nitrogen, certain extreme conditions can trigger a series of reactions from which the nitrogen in the atmosphere is involved in the production of nitric acid, causing acid rain.

Step 1:	The temperatures reached inside a combustion engine provide the high activation required to overcome the high bond enthalpy of nitrogen, allowing the formation of nitrogen(II) oxide
Step 2:	Nitrogen(II) oxide further reacts to give nitrogen(IV) oxide
Step 3:	Nitrogen(IV) oxide dissolves in water and reacts with more oxygen to form dilute nitric acid

i)
Write **three** equations to show each of these reactions. Include state symbols.

[3]

ii)
Identify the changes in oxidation number of nitrogen in each reaction and use this to state if nitrogen is oxidised or reduced in each reaction.

[3]

[6 marks]

Question 3c

A car travelled one kilometre and released 0.23 g of an oxide of nitrogen, N_xO_y, which occupies 120 cm^3.

Calculate the values of x and y.
(Assume 1 mol of gas molecules occupies 24.0 dm^3).

[3 marks]

Model Answers: Hard

1a

a) An equation to show how NO_2 is formed in these situations is:

- $N_2 + 2O_2 \rightarrow 2NO_2$
  OR
  $2NO + O_2 \rightarrow 2NO_2$; [1 mark]

[Total: 1 mark]

- You can either give the equation to show the formation of nitrogen dioxide from nitrogen and oxygen present in the atmosphere or show the oxidation of nitrogen monoxide

1b

b)

i) NO_2 is removed from the exhaust gases of motor vehicles by:

- A catalytic converter
  AND
  Passing the exhaust gases over a catalyst / Pt / Rh; [1 mark]

ii) An equation for this process is:

- $NO_2 + 2CO \rightarrow \frac{1}{2} N_2 + 2CO_2$;
  OR
  $2NO_2 + CH_4 \rightarrow CO_2 + N_2 + 2H_2O$; [1 mark]

[Total: 2 marks]

- **Careful**: Sometimes you just need to state that a catalytic converter is used to remove NO_2 from exhaust gases, but other times, such as in this question, you are also required the give the extra detail explaining that the exhaust gases are passed over a catalyst, you are best to give this information just in case!

- The expected equation for this reaction is with carbon monoxide to produce nitrogen and carbon dioxide
    - The other equation is also acceptable although you are not expected to know this

1c

c) Suggesting if NO_2 would be reduced if hydrogen was used as a fuel for combustion:

- No, it wouldn't be reduced
  AND
  Because the reaction does not require a particular fuel / NO_2 is formed from N_2 and O_2 in air during combustion; [1 mark]

[Total: 1 mark]

- You need to give an explanation to gain the mark, just stating it wouldn't be reduced is insufficient
- The fuel used for combustion has no impact on the reaction
- Nitrogen and oxygen used to form nitrogen dioxide both come from the air, not the fuel, and it is the high temperatures found in the combustion engine which allow them to react

1d

d)

i) The environmental significance of this reaction is:

- SO_3 produces acid rain; [1 mark]

ii) An equation to show how the NO_2 is regenerated in the second step of the oxidation is:

- $NO + 1/2\ O_2 \rightarrow NO_2$; [1 mark]

iii) The position of the equilibrium:

- Will shift to the right as height increases; [1 mark]
- Because the reaction is exothermic; [1 mark]

[Total: 4 marks]

- When SO_2 is oxidised, it forms SO_3 which reacts with rainwater to form dilute sulfuric acid as follows:
  - $SO_3 (g) + H_2O (l) \rightarrow H_2SO_4 (aq)$
- You need to know both equations which show how NO_2 catalyses this reaction
  - NO_2 is used in reaction **1** which is the first step and regenerated in the second step
- When explaining how equilibrium is affected with changing conditions, you must always state in which direction equilibrium will shift AND why
- As you move further from the Earth's surface, i.e. height increases, the temperature decreases
- A decrease in temperature will always favour the exothermic reaction, in this case, this is the forward direction, as the ΔH value is negative
  - The yield of NO and SO_3 increases
- The reverse argument would be accepted, i.e. equilibrium shifts to the left as height decreases

2a

a)

i) Ammonia can be prepared from ammonium chloride and calcium hydroxide by:

- Mixing and heating (ammonium chloride and calcium hydroxide); [1 mark]
- Drying (the ammonia) with calcium oxide; [1 mark]

ii) The presence of ammonia gas can be confirmed as:

- It turns damp red litmus paper blue; [1 mark]

[Total: 3 marks]

- Ammonium chloride and calcium hydroxide are both solids
- The gas formed in the reaction is typically passed into a U-tube containing calcium oxide, which absorbs any water present
- You should know the test for ammonia gas

2b

b) The reaction between ammonium chloride and calcium hydroxide is an example of an acid-base reaction because:

- NH_4^+ acts as an acid as it is a proton donor / it donates a proton; [1 mark]
- OH^- acts as a base as it is a proton acceptor / it accepts a proton; [1 mark]

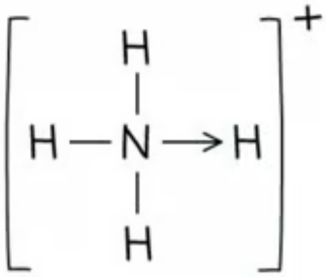
[Total: 2 mark]

- You need to identify which species is acting as an acid and as a base and also explain why they are an acid and base in terms of protons

2c

c)

i) A dot-and-cross diagram of the ammonium ion showing all of the bonds within the molecule is:



- Arrow pointing from N to a H atom; [1 mark]
- Rest of structure including square brackets and + charge; [1 mark]

ii) The N-H bonds being the same length means that:

Any **one** from the following:

- All four covalent bonds are equivalent; [1 mark]
- The positive charge is spread evenly around the ion; [1 mark]

[Total: 3 marks]

- The ammonium ion has a coordinate bond which is shown by an arrow that points from the atom that is donating the lone pair of electrons (N atom) to the atom that is accepting the lone pair (H^+ ion)
- Once the coordinate bond is formed, the bond is indistinguishable from the other

is accepting the lone pair (H^+ ion)
- Once the coordinate bond is formed, the bond is indistinguishable from the other bonds

3a

a)

i) Another reason why the nitrogen molecule is so unreactive is:

- It is non-polar; [1 mark]

ii) The type of bonds that exists between the two nitrogen atoms are:

- One σ / sigma bond; [1 mark]
- Two π / pi bonds; [1 mark]

[Total: 3 marks]

- As nitrogen is made from two nitrogen atoms, the electrons are shared equally and the molecule is non-polar
- This means that it is not attracted to or likely to react with other molecules
- The triple covalent bond is made from one sigma bond which is due to the end of overlap of hybridised sp orbitals and two pi bonds, due to the sideways overlap of p orbitals
- The two pi orbitals are at right angles to each other

3b

b)

i) **Three** equations to show each of these reactions are:

- Step 1: $N_2 (g) + O_2 (g) \rightarrow 2NO (g)$; [1 mark]
- Step 2: $2NO (g) + O_2 (g) \rightarrow 2NO_2 (g)$; [1 mark]
- Step 3: $2NO (g) + H_2O (l) + ½O_2 (g) \rightarrow 2HNO_3 (aq)$; [1 mark]

ii) The changes in oxidation number of nitrogen in each reaction and stating if nitrogen is oxidised or reduced in each reaction:

- Step 1: 0 to +2 **AND** oxidation; [1 mark]

- Step 2: +2 to +4 **AND** oxidation; [1 mark]
- Step 3: +4 to +5 **AND** oxidation; [1 mark]

[Total: 6 marks]

- Nitrogen in nitrogen(II) oxide has the oxidation +2, as indicated by the Roman numeral
  - Its formula is NO
    - Oxygen has an oxidation number of -2, therefore there must be one nitrogen with an oxidation number of +2 so that the overall oxidation number of the compound is 0
  - It is produced by the reaction between nitrogen and oxygen in the atmosphere
- Nitrogen in nitrogen(IV) oxide has the oxidation +4, as indicated by the Roman numeral
  - Its formula is NO_2
    - Oxygen has an oxidation number of -2, therefore there must be two oxygen atoms (2 x -2 = -4) and one nitrogen with an oxidation number of +4 so that the overall oxidation number of the compound is 0
  - It is produced by the reaction between nitrogen(II) oxide and oxygen in the atmosphere
- Nitrogen in nitric acid has the oxidation +5, which you can work out from its formula
  - Its formula is HNO_3
    - Oxygen has an oxidation number of -2 and there are 3 oxygen atoms: 3 x -2 = -6
    - Hydrogen has an oxidation number of +1
    - Therefore oxidation number of nitrogen must be +5 so that the overall oxidation number of the compound is 0
  - It is produced by the reaction between nitrogen(II) oxide, water oxygen in the atmosphere
- In each reaction, the oxidation number of nitrogen increases, therefore nitrogen is being oxidised in each reaction

3c

c) The values of *x* and *y* are:

- Moles of $N_xO_y = \frac{120}{24\,000}$ = 0.005 (mol); [1 mark]

- $M_r = \frac{\text{mass}}{\text{moles}} = \frac{0.23}{0.005}$ = 46 (g mol^{-1}); [1 mark]
- x = 1 **AND** y = 2; [1 mark]

[Total: 3 marks]

- To calculate the number of moles of N_xO_y, make sure that both the volume of the gas and the molar gas volume are both in either cm^3 or dm^3, so the calculation is either 120 ÷ 24,000 or 0.12 ÷ 24
- Once you have deduced the M_r of N_xO_y, it can be a bit of trial and error to find the number of oxygen and nitrogen atoms:
- You know that it contains at least 1 atom of nitrogen = 14 g mol^{-1} and 1 atom of oxygen = 16 g mol^{-1}
  - 14 + 16 = 30
- This leaves 16 g mol^{-1} unaccounted for, which means there must be another oxygen atom
  - 14 + (2 x 16) = 46